

1. Routing

Built a POML prompt that classifies a question into `python_book`, `java_book`, `javascript_book`, or clarify if the language is ambiguous.

The prompt uses a chain-of-thought instruction block internally but is told to output JSON only so it's cheap to parse.

`route_question()` in `rag_pipeline.py` sends this prompt to `gpt-oss-20b` at `temperature=0` for deterministic classification, then extracts the JSON with a regex fallback in case the model wraps its answer in extra text.

2. Retrieval

`query_collection()` queries the routed ChromaDB collection returning the top 5 chunks with their content, metadata, and L2 distance.

Both ingestion and query time use the same embedding function with `all-MiniLM-L6-v2` so query and document vectors live in the same space.

3. Answering

`answer.poml` implements four prompting techniques as conditional blocks, all sharing the same injection:

Zero-Shot

Few-Shot - two examples

Chain of Thought - explicit reasoning steps, output split into reasoning and answer sections

Advanced (Self-Consistency) independently reasons through the question three separate times, each grounded only in the retrieved context, then compares the three conclusions and presents whichever answer the majority agree on.

`generate_response()` renders the selected technique's prompt and calls ChatGroq at `temperature=0.3`

4. Comparison & Judging

multi-model and multi-technique each route and retrieve once, then fan the same context out to multiple ChatGroq calls.

`judge.poml` takes the question, comparison type, and a labeled list of responses, and returns strict JSON (rankings, feedback, summary) graded on correctness, groundedness,

clarity, and completeness. The judge runs on gpt-oss-120b at temperature=0 for consistent grading.

Dev Takeaways

Chunking

Started at chunk_size of 100, which was too small. It split the Java PDF into 11,600 chunks and went past ChromaDB's per-call batch limit of 5,461 and crashing ingestion.

Moved to chunk_size=1000 which fixed the batch size issue and produced more semantically complete chunks

Retrieval

L2 distance scores in the 0.30–0.42 range corresponded to genuinely relevant chunks

Retrieval quality is very sensitive to how book like the PDFs are. Chunks from history sometimes outrank more technical chunks for broad questions like "tell me about Java," which shaped the answer's content.

Model Selection

Router model: gpt-oss-20b for a small and fast is sufficient for a simple 4 way classification task, keeping routing latency low.

Judge model: gpt-oss-120b is the largest available model for grading, since judgment quality benefits more from model size than generation does.

In the Python few-shot comparison below, interestingly the smallest-context judge ranking didn't correlate with model size.

Qwen3 27B outranked both GPT-OSS models. This suggests bigger isn't automatically better for a given prompting technique/task combo, and answer length/structure mattered as much as raw model capability to the judge.

Prompting Techniques

Chain of Thought visibly produced a structured Reasoning

Few-Shot (Python test below) produced consistently well-formatted, structured answers across all three models without needing explicit step by step instructions

Test Results

Test 1: Single Model & Single Prompting Technique (Chain of Thought)

Question: "Tell me about Java"

Routing: java_book

Result: Retrieved 5 chunks (distances 0.30–0.42) from the Java book's preface/intro, and the CoT prompt produced a clearly separated reasoning and answer. Output grounded in Gosling's quotes and the book's package coverage (java.lang, java.util.concurrent, etc.).

```
=====
Select Response Mode
=====
1. Single Model & Single Prompting Technique
2. Compare Multiple Models using Single Prompting Technique
3. Compare Multiple Prompting Techniques using Single Model
=====

Enter choice: 1

=====
Select Prompting Technique for Response Generation
=====
1. Zero-Shot
2. Few-Shot
3. Chain of Thought
4. Advanced (Chain-of-Verification)
=====

Enter choice: 3

-----
Enter your question: Tell me about Java
-----

RESULTS
=====

[Routing]
  Routed to: java_book

[Retrieved Context] (5 chunks)

--Chunk 15 (collection: java_book, distance: 0.3844):
  Gosling, and Holmes, or The Java™ Language Specification by Gosling, Joy, yours
  truly, and Bracha. ...

--Chunk 32 (collection: java_book, distance: 0.3332):
  --- Page 22 ---
  1
  CHAPTER 1
  Introduction
  THIS book is designed to help you make effective use of the...

--Chunk 20 (collection: java_book, distance: 0.3875):
  twenty-three items, but I thoroughly revised all the original material and retired a
  few items whose...

--Chunk 16 (collection: java_book, distance: 0.3950):
  "blue collar language" that was "pretty simple" [Gosling97]. At about the same time,
  Bjarne Stroustr...

--Chunk 21 (collection: java_book, distance: 0.4216):
  In 1996 I pulled up stakes and headed west to work for JavaSoft, as it was then
  known, because it wa...

-----
[Answer]
-----
**Reasoning:**
1. Identify the request: the user wants a general description of Java.
2. Relevant excerpts:
  - Preface notes that Java was described by James Gosling as "pretty simple" and later grew in size and complexity.
  - The introduction states that the book focuses on the core libraries 'java.lang', 'java.util', 'java.io', and sub-packages such as 'java.util.concurrent' and 'java.util.function'.
  - The historical note mentions Java's rapid evolution, the addition of many features in releases 5 and 6, and the current state of a large, multi-paradigm platform.
3. Combine these points into a concise description that stays within the provided text.

**Answer:**

Java is a programming language created by James Gosling (often called "the father of Java"). In its early days it was praised for being "pretty simple" – a language with a clean, straightforward design that made it easy for newcomers to learn and for experienced developers to write reliable code.

Over the years Java has grown dramatically. New language features and a rich set of libraries were added, especially in releases 5 and 6, turning the platform into a large, complex ecosystem. Today Java includes a broad core library set- 'java.lang', 'java.util', 'java.io' –and many sub-packages such as 'java.util.concurrent' for concurrency and 'java.util.function' for functional-style programming.

Because of this evolution, Java now supports multiple programming abstractions (e.g., object-oriented, functional, and concurrent paradigms) and provides extensive APIs for everything from collections to I/O, making it a powerful but also more intricate language than its original "simple" incarnation.
```

Test 2: Compare Multiple Models (Few-Shot)

Question: "Tell me about Python"

Result: All three models (Qwen3 27B, GPT-OSS 20B, GPT-OSS 120B) produced grounded, well-structured answers. The LLM judge ranked Qwen3 27B first for completeness, GPT-OSS 20B second, and GPT-OSS 120B last for being too brief despite being accurate.

```
Code Documentation Chatbot

=====
Select Response Mode
=====
1. Single Model & Single Prompting Technique
2. Compare Multiple Models using Single Prompting Technique
3. Compare Multiple Prompting Techniques using Single Model
=====

Enter choice: 2

=====
Select Prompting Technique for Response Generation
=====
1. Zero-Shot
2. Few-Shot
3. Chain of Thought
4. Advanced (Chain-of-Verification)
=====

Enter choice: 2

-----
Enter your question: Tell me about Python

=====
MODEL COMPARISON
=====

🌟 Qwen3 27B

Based on the provided documentation, here is an overview of Python:

**Definition and Nature**
Python is a simple, easy-to-understand, high-level, and interpreted programming language. It is designed to feel like reading simple English (pseudo-code), making it accessible for beginners. It is used to instruct computers to perform various tasks.

**Key Features**
* **Simplicity:** Easy to understand, which leads to less development time.
* **Open Source:** It is free and open source.
* **Portable:** It works across multiple operating systems, including Linux, Windows, and Mac.
* **Versatile:** It supports a broad range of libraries and frameworks, making it suitable for web development, data analysis, AI, and more.
* **User-Friendly:** It is considered fun to work with and is an ideal starting point for those new to coding.

**Installation**
Python can be easily installed by downloading it from python.org. After clicking the download button and completing the setup for your specific platform, the installation is ready to use.

**Basic Usage**
* **Writing Code:** You can create a file (e.g., `hello.py`) and write code such as `print("hello world")`.
* **Execution:** You can execute the file by typing `python hello.py` in the terminal.
* **Calculator Mode:** You can use Python as a calculator by typing `python` and pressing Enter in the terminal.
* **Modules:** Python uses modules (files containing code) to extend functionality. There are built-in modules (like `os` and `random`) and external modules (like `flask` and `tensorflow`) which can be installed using the package manager pip (e.g., `pip install flask`).
```

```

🔥 GPT-OSS 20B

**Python – a quick overview**

- **What it is**
  Python is a "high-level, interpreted" programming language that feels like reading simple English. Its pseudo-code style makes it easy to learn and understand, especially for beginners.

- **Key features**
  - **Easy to understand** – less development time.
  - **Free and open source** – available on Linux, Windows, and macOS.
  - **Portable** – runs on all major operating systems.
  - **Versatile** – used for web development, data analysis, AI, and many other domains.

- **Installation**
  Download the installer from [python.org](https://python.org). After clicking the download button, run the installer for your platform; the setup completes automatically.

- **Getting started**
  1. Create a file, e.g., `hello.py`.
  2. Add a simple program:
     ```python
 print("hello world")
     ```
  3. Run it from the terminal: `python hello.py`.
     The output will be `hello world`.

- **Modules & pip**
  - A "module" is a file containing reusable code.
  - Python ships with "built-in modules" (e.g., `os`, `random`).
  - "External modules" must be installed with the package manager "pip":
     ```bash
 pip install flask
     ```
     This installs the Flask web framework, for example.

- **Using Python as a calculator**
  Simply start the interpreter by typing `python` in a terminal and you can perform calculations interactively.

- **Audience & purpose**
  The handbook is aimed at beginners, students, professionals, and enthusiasts. It covers fundamentals, practical examples, exercises, and references to official documentation for deeper learning.

In short, Python is a beginner-friendly, powerful language that's widely used across many fields and runs on almost any platform.

🔥 GPT-OSS 120B

**What Python is**

- Python is a simple, easy-to-understand language that reads almost like plain English. Its "pseudo-code" style makes it especially friendly for beginners [Page 7] .

**Key characteristics**

- **Easy to understand** → reduces development time.
- **Free and open source**.
- **High-level, interpreted language** with a broad ecosystem of libraries and frameworks.
- **Portable** – runs on Linux, Windows, and macOS.
- **Versatile** – used for web development, data analysis, AI, and many other domains [Page 2] .

**Why it's popular**

- Simplicity and readability make it ideal for newcomers, yet its power and extensive libraries keep it valuable for experienced programmers as well [Page 2] .

=====
LUM JUDGE EVALUATION
=====

RANKINGS:
1st: Qwen3 27B
2nd: GPT-OSS 20B
3rd: GPT-OSS 120B

FEEDBACK:

Qwen3 27B:
  Provides a thorough, accurate overview with clear sections on definition, features, installation, basic usage, and module management, making it the most complete and easy to follow.

GPT-OSS 20B:
  Also accurate and well-structured, but slightly less detailed than Qwen3 27B (e.g., fewer explicit installation steps) while still covering key points.

GPT-OSS 120B:
  Correct but overly brief; lacks practical details such as installation instructions and code examples, reducing its usefulness and completeness.

```

Test 3: Self consistency

Question: "Tell me about Java"

```
Enter your question: Tell me about java

=====
RESULTS
=====

[Routing]
  Routed to: java_book

[Retrieved Context] (5 chunks)

--Chunk 15 (collection: java_book, distance: 0.3044):
  Gosling, and Holmes, or The Java™ Language Specification by Gosling, Joy, yours
  truly, and Bracha. ...

--Chunk 32 (collection: java_book, distance: 0.3332):
  --- Page 22 ---
1
CHAPTER 1
Introduction
THIS book is designed to help you make effective use of the...

--Chunk 20 (collection: java_book, distance: 0.3875):
  twenty-three items, but I thoroughly revised all the original material and retired a
  few items whose...

--Chunk 16 (collection: java_book, distance: 0.3950):
  "blue collar language" that was "pretty simple" [Gosling97]. At about the same time,
  Bjarne Stroustrup...

--Chunk 21 (collection: java_book, distance: 0.4216):
  In 1996 I pulled up stakes and headed west to work for JavaSoft, as it was then
  known, because it wa...

-----
[Answer]
-----

**Path 1:**
The Java programming language was created by James Gosling (often called "the father of Java"). Early on it was described as a "blue-collar language" that was "pretty simple." Over the years the platform has grown dramatically, adding many abstractions (parallel execution, iteration, dates-and-times, etc.) and becoming "large and complex." Its core libraries are the packages java.lang, java.util, and java.io, together with sub-packages such as java.util.concurrent and java.util.function.

**Path 2:**
Java is presented in the book as a language whose fundamental libraries (java.lang, java.util, java.io and related sub-packages) are the basis for effective programming. The book's purpose is to teach "customary and effective usage" of these libraries, showing patterns and idioms that make Java programs "pleasant, elegant, and graceful" and that help developers avoid headaches when modifying code.

**Path 3:**
Both James Gosling and Bjarne Stroustrup noted that Java started out relatively simple, but "as time passes, Java will grow significantly in size and complexity." Today it indeed "has multiple abstractions for many things, from parallel execution, to iteration, to the representation of dates and times," confirming that the language has expanded far beyond its original simplicity.

**Answer:**
Java is a programming language originally designed by James Gosling to be simple and approachable ("blue-collar language"). Since its inception it has expanded into a large, complex platform that provides a rich set of core libraries—java.lang, java.util, java.io, and many sub-packages such as java.util.concurrent and java.util.function—covering everything from basic utilities to advanced concurrency and functional programming. The book referenced in the excerpts aims to teach developers how to use these libraries effectively, emphasizing good design patterns, idioms, and practices that keep Java code clear, maintainable, and enjoyable to work with.

=====
```

Result: Self Consistency routed correctly to java_book and generated the three different reasoning paths. It converged on a single final answer correctly showing genuine cross checking.